

Jordan Cahoon

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Education

2024-2029	STANFORD UNIVERSITY PhD in Biomedical Data Science GPA: 4.0
2020-2024	UNIVERSITY OF SOUTHERN CALIFORNIA BS in Computer Science, Minor in Computational Biology & Bioinformatics GPA: 3.86

Research Experience

Jun 2025 - Present	PhD Student. ADVISED BY EMILY ALSENTZER & STEPHEN MONTGOMERY Stanford University, Department of Biomedical Data Science • Building a benchmark dataset to evaluate information retrieval performance in electronic health records, supporting long-context clinical reasoning in large language models. • Developing computational methods that integrate electronic health records and multi-omics data to accelerate rare disease diagnosis.
Feb 2021 - Jun 2024	Undergraduate Researcher. ADVISED BY CHARLESTON CHIANG University of Southern California, Keck School of Medicine • Conducted population genetics research aimed at improving generalizability across ancestrally diverse populations. • Identified discrepancies in genotype imputation accuracy for non-European populations (e.g., East Asian, South Asian, Oceanian, Southeast Asian) using the TOPMed reference panel. • Designed a framework using meta-imputation that improved imputation accuracy for East and Southeast Asian cohorts, with enhanced recovery of population-specific variants. • Developed a deep learning model leveraging genealogical trees to detect archaic introgression, achieving robust performance across diverse demographic scenarios.
Oct 2022 - Sep 2023	Undergraduate Researcher. ADVISED BY LUIS A. GARCIA University of Southern California, Department of Computer Science • Evaluated generalized stress detection for health workers using wearable devices across three datasets with shared stress representations. • Demonstrated gradient boosting trees outperform linear support vector machines, random forest, and feed-forward neural networks for continuous stress detection.

Jun 2020 -
Jan 2021

Undergraduate Researcher.

ADVISED BY ELIA TAIT-WOJNO

University of Washington, Department of Immunology

- Elucidated how the PGD2-CRTH2 pathway suppresses Type 2 intestinal immune response during helminth infections in murine models with single-cell RNA sequencing analysis.
- Identified canonical markers of epithelial immune cell types and characterized gene expression in inflammatory immune response in murine cecum during infection.

Jul 2019 -
Aug 2019

Research Intern.

ADVISED BY JAKE VALENZUELA

The Institute for Systems Biology, Baliga Lab

- Developed electroporation protocol that facilitates the transfer of CRISPR-Cas9 complex into *C. reinhardtii*.
- Created and presented how nitrogen starvation increases lipid production in *C. reinhardtii*.

Work Experience

Apr 2025 -
Present

Program Lead

STANFORD SUMMER RESEARCH PROGRAM

- Led program development and coordination for undergraduate research interns from underrepresented institutions, with a focus on graduate school preparation.
- Designed and taught an introductory programming workshop for wet lab researchers, introducing core data science concepts.

Jun 2021 -
July 2023

Viterbi Student Ambassador, Content Lead.

USC VITERBI SCHOOL OF ENGINEERING ADMISSIONS

- Led team of 12 students to produce bi-weekly virtual student panels about student life for audiences of 200+ prospective engineering students.
- Advertised panels through social media outreach, bi-weekly YouTube videos, and Spotify podcasts.

Aug 2022 -
May 2023

Artificial Intelligence Intern.

THE ELLISON INSTITUTE FOR TRANSFORMATIVE MEDICINE

- Developed deep learning models to automate the diagnosis for breast and prostate cancer from digital pathology.
- Refined quality control pipeline to process thousands of whole slide images (WSI) in the cloud.

May 2022 -
Aug 2022

Software Engineering Intern.

ORACLE CLOUD INFRASTRUCTURE

- Designed and tested automated daily health checks for cloud billing accounts.

Awards & Honors

Warren Alpert Computational Biology & Artificial Intelligence Fellowship 2025
USC Discovery Scholar 2024
USC Google Alumni Award in Quantitative Biology 2024
Knight-Hennessy Scholarship Finalist 2024
Barry Goldwater Scholarship 2023
Astronaut Scholarship Finalist 2023
USC Viterbi Dean's List 2020-2024
USC Viterbi Undergraduate Merit Research Fellowship 2020
USC Presidential Scholarship 2020
USC Dornsife Thematic Option, Reading & Writing 2020

In Preparation

Cahoon JL, Stanwyck CO, Somani S, Chung P, Keet KR, Black KC, Fisher AT, Khemani S, Liu J, Ma S, Maharaj SK, Pandya RM, Perez-Guerrero E, Pillai P, Shieh L, Wu DJH, Xie J, McAvoy JC, Nguyen T, Tran J, Yin L, Lin B, Callahan A, Fries JA, Shah NH, Alsentzer E. *"A Living Benchmark for Information Retrieval from Electronic Health Records."* Manuscript in preparation.

Ungar RA, **Cahoon JL**, Reuter C, Carter J, Bonner D, Cho M, Montgomery SB, Alsentzer E, Halley M. *"Impact of participant-, clinician-, and EHR-derived phenotyping on rare disease diagnosis: a mixed-methods study."* Manuscript in preparation.

Publications & Preprints

- [8] Ronaghi S, Tonekaboni S, Stempfle L, Utti V, **Cahoon JL**, Hendrix N, Vala A, Ghassemi M, Alsentzer E. *"Clinically Grounded Privacy Evaluation of Medical LMs."* arXiv preprint. 2026 Jun 8. [Accepted at EMNLP 2026]
doi: 10.48550/arXiv.2606.09590.
- [7] **Cahoon JL**, Stanwyck C, Aali A, Madding R, Sun E, Jiang Y, Dhanasekaran R, Alsentzer E. *"Clinical Note Bloat Reduction for Efficient LLM Use."* arXiv preprint. 2026 Mar 21. [In Review]
doi: 10.48550/arXiv.2604.16364. .
- [6] Aali A, Mohsin MA, Bikia V, Singhvi A, Gaus R, Bedi S, Cui H, Fuentes M, Unell A, Mai Y, **Cahoon JL**, Pfeffer M, Daneshjou R, Koyejo S, Alsentzer E, Potts C, Shah NH, Chaudhari AS. *"Structured Prompts Improve Evaluation of Language Models."* arXiv preprint. 2025 Nov 25. [In Submission]
doi: 10.48550/arXiv.2511.20836.
- [5] Fan C, **Cahoon JL**, Dinh BL, Vecchy DO, Huber C, Edge MD, Mancuso N, Chiang CWK. *"A likelihood-based framework for demographic inference from genealogical trees."* Nature genetics. 2025 Mar 20.
doi: 10.1038/s41588-025-02129-x.
- [4] **Cahoon JL**, Rui X, Tang E, Simons C, Langie J, Chen M, Lo YC, Chiang CWK. *"Imputation Accuracy Across Global Human Populations."* American Journal of Human Genetics. 2024 May 2. [International Genetic Epidemiology Society 32nd Annual Meeting Best Poster]
doi: 10.1016/j.ajhg.2024.03.011.
- [3] **Cahoon JL**, Garcia L. *"Continuous Stress Monitoring for Healthcare Workers: Evaluating Generalizability Across Real-World Datasets."* The 14th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB). 2023 Sept 3.
doi: 10.1145/3584371.36129741.
- [2]

Sheng X, Xia L, **Cahoon JL**, Conti DV, Haiman CA, Kachuri L, Chiang CWK. "Inverted genomic regions between reference genome builds in humans impact imputation accuracy and decrease the power of association testing." Human Genetics and Genomics Advances. 2022 Nov 11. [HGGAdvances Award for Outstanding Early Career Publication]

doi: 10.1016/j.xhgg.2022.100159.

- [1] Oyesola OO, et. al. "PGD2 and CRTH2 counteract Type 2 cytokine-elicited intestinal epithelial responses during helminth infection." Journal of Experimental Medicine. 2021 Sep 6;218(9):e20202178. doi: 10.1084/jem.20202178.

Oral Presentations & Other Projects

- May 2025 Self-Guided Query Expansion for Longitudinal Clinical Retrieval, *Biomedical Data Science Student Seminars*, Stanford University.
- May 2025 Impact of Sanctuary Policies on Healthcare Utilization Among U.S. Agricultural Workers: A Causal Inference Approach, *Outcomes Analysis Final Presentation*, Stanford University.
- May 2023 Imputation Efficacy Across Global Human Populations, *Southern California Evolutionary Genetics and Genomics Meeting 2023*, University of California Irvine.
- Dec 2022 Assessing Imputation Quality for Diverse Populations, *Department Research Seminar, Center for Genetic Epidemiology* Keck School of Medicine.
- Nov 2022 Predicting Foster Care Outcomes in the United States with the National Youth in Transition Database, *Artificial Intelligence for Sustainable Development Final Presentation*, University of Southern California.

Teaching

- 2025 - 2026 Teaching assistant for Representations and Algorithms for Computational Molecular Biology (BMDS 214)
- Jun - Aug 2025 Stanford Summer Research Program Introduction to Programming
- Jan - Apr 2023 Project mentor for undergraduate machine learning projects through CAIS++
- Sep - Dec 2022 Curriculum Lead for open source deep learning curriculum for undergraduates
- Jan - May 2022 Course Producer for CSCI 104, Data Structures and Objected Oriented Programming

Mentorship

Venkat Ramaraju (ICME Masters, Alsentzer Lab, 2026)
 Lucy Yin (DBDS Masters, Alsentzer Lab, 2026)
 Alan Zhao (Undergraduate, SSRP 2025)
 Kamille See (Undergraduate, SSRP 2025)

Leadership

- May 2026 **Organizer** Warren Alpert Computational Biology and AI Research Symposium.
- Aug 2022 - **Co-President.** The Center for Artificial Intelligence's Student Branch (CAIS++)
- May 2023 Directed all organization initiatives, including 5-8 semester projects, 4 Fall curriculum groups, two project showcases, weekly general meetings, and speaker events to engage undergraduates in artificial intelligence.
- Sep 2021 - **Project Manager.** Novus Think Tank. Oversaw six focus project groups that targeted critical social issues impacting the university and surrounding areas.
- May 2022